

Aspect-Based Stance and Framing Analysis of News

A Two-Stage Sentence–Aspect and Article-Level Classification System

Detailed Project Report

Dataset, Annotation, Architecture, Experiments, Results, and User Interface

Project objective: identify which governmental aspect a Bangla news sentence discusses, determine whether its framing is neutral, favorable, or critical toward the government, and aggregate these interpretable decisions into an overall article judgment. The deployed interface additionally compares two articles reporting the same event.

Bangla News Framing Project

September 2026

Abstract

This project develops an interpretable natural-language-processing pipeline for analysing governmental framing in Bangla news. The project uses the publicly available BanglaBias dataset as its base article corpus [2, 3]. The original BanglaBias resource provides Bangla news articles and article-level stance labels; however, the sentence-level aspect-based task developed in this project is different from the original benchmark task. We therefore further processed and annotated a subset of the base articles at sentence level using six governmental aspects and four sentence–aspect labels.

The resulting system does not merely assign one opaque label to an article. Instead, every sentence is paired with six predefined aspects and processed by two BanglaBERT classifiers. Stage 1 determines whether the sentence is relevant to an aspect, and Stage 2 classifies each relevant pair as neutral (N), government-favouring (GF), or government-critical (GC). Pair probabilities are aggregated into twelve article features—six aspect-emphasis values and six directional stance values—which are passed to a balanced multinomial logistic-regression classifier for the final article label. The system is exposed through a local Gradio interface supporting both single-sentence inspection and comparison of two reports about the same event.

On the protected event-held-out test split, Stage 1 achieved 0.9631 accuracy and 0.9457 macro F1. Stage 2 achieved 0.9348 accuracy but only 0.5098 macro F1 because rare GF and GC examples remained difficult. The complete four-label sentence–aspect pipeline achieved 0.9504 accuracy and 0.6060 macro F1. The article classifier reached 0.4444 accuracy and 0.4254 macro F1 on 18 held-out articles, exceeding the neutral-majority macro F1 of 0.1449 but failing to recover any neutral article. Experiments with stronger class weighting and an exactly balanced 2,940-row Stage 2 dataset were carefully evaluated. Although the latter reached 0.9224 macro F1 on its internal selection validation set, it fell to 0.4375 macro F1 on the protected original-news test set and reduced end-to-end macro F1 to 0.5455. Consequently, the original Stage 2 model remains the selected final model. These results show both the value of the staged design and the importance of event-level, source-like evaluation under label scarcity.

1 Introduction and Actual Task

News framing concerns the selection and emphasis of particular aspects of an issue in a way that influences how it is understood [1]. In this project, the practical task is to analyse Bangla-language news reporting about government-related events at three connected levels:

1. **Sentence–aspect relevance:** for each sentence and each of six governmental aspects, decide whether the sentence discusses that aspect.
2. **Sentence–aspect stance:** for every relevant pair, determine whether the wording is neutral, government-favouring, or government-critical.
3. **Article-level framing:** aggregate the sentence decisions into interpretable aspect features and predict an overall article label: *Govt leaning*, *Neutral*, or *Govt critique*.

The same-event comparison task then applies the same pipeline independently to two articles and displays their overall labels, confidence, aspect emphasis, aspect stance, and the differences between them. This makes the model suitable for a classroom demonstration in which a user may enter one sentence, a complete article, or two reports about the same event.

The task is deliberately narrower than general-purpose bias detection. The output represents learned textual framing signals under the project annotation scheme. It does not establish factual truth, political intention, journalistic quality, or whether a publisher is inherently biased.

2 Label Scheme and Interpretable Aspects

Each sentence is evaluated against all six aspects in Table 1. Full descriptive aspect text, rather than only the short aspect name, is paired with the sentence so that the transformer receives the intended semantic definition.

Table 1: The six governmental aspects used throughout annotation, training, aggregation, and the UI.

Code	Aspect	Scope represented by the aspect description
A1	Policy & Administration	Government policy, administrative decisions, implementation, public institutions, and delivery of state functions.
A2	Accountability & Justice	Responsibility, corruption, investigation, courts, punishment, transparency, and demands for justice.
A3	Security & Civil Rights	Law and order, public security, violence, policing, freedom, human rights, and civil liberties.
A4	Democracy & Mobilization	Elections, political participation, protest, rallies, opposition activity, representation, and democratic processes.
A5	Economy & Development	Economic conditions, public expenditure, infrastructure, employment, trade, growth, and development projects.
A6	Welfare, Social Issues & Environment	Relief, health, education, vulnerable groups, social welfare, disasters, climate, and environmental effects.

The four pair-level labels are:

- NR (not relevant): the sentence does not express the target aspect;
- N (neutral): it is relevant but gives no clearly favourable or critical governmental framing;
- GF (government-favouring): the relevant sentence presents the government, its action, or its performance favourably; and
- GC (government-critical): the relevant sentence criticises governmental action, omission, responsibility, or performance.

The staged formulation is important: NR is fundamentally a relevance decision, while N/GF/GC are stance decisions conditional on relevance. A single four-class model would ask one classifier to learn both problems simultaneously while the dominant NR class overwhelms the rare directional classes.

3 Data and Annotation

3.1 Base dataset and raw article corpus

The base corpus for this project was obtained from the publicly available **BanglaBias** dataset hosted on Hugging Face [3]. BanglaBias accompanies the work “*Read Between the Lines: A Benchmark for Uncovering Political Bias in Bangla News Articles*” [2]. The dataset was created for article-level political stance and bias analysis in Bangla news. It contains 200 Bangla news articles and assigns each complete article one of three stance labels: *Govt critique*, *Neutral*, or *Govt leaning*.

The original BanglaBias dataset and the task performed in this project should therefore be distinguished clearly. **BanglaBias serves as the base article dataset, but its original task is not identical to the aspect-based sentence-level task developed in this project.** The original dataset provides complete articles, metadata, events, sources, and article-level stance labels. It does not provide the six sentence-level governmental aspect annotations (A1--A6) used by our two-stage relevance-and-stance system. Those finer-grained annotations were introduced later as part of this project.

The raw BanglaBias corpus contains 200 Bangla news articles covering 46 events and 54 news sources. Its eight fields are `id`, `event`, `news_headline`, `news_body`, `source_link`, `date`, `news_corpora_name`, and the article-level `label1`. The original label distribution in the project copy is 95 *Govt critique*, 72 *Neutral*, and 33 *Govt leaning*. The Hugging Face dataset page reports 200 rows and the same article-level data structure [3].

Figure 1 is a direct visual sample generated from the raw Parquet file; long text is only visually shortened in the screenshot and is not altered in the source dataset.

Raw Bangla News Corpus — Article-Level Sample						
Three example records from the 200-article source corpus (46 events, 8 original fields)						
id	event	news_headline	news_body (excerpt)	source	date	article label
1	তাজরীন ফাশন অরিকাত	তাজরীন অরিকাত: 'কেয়ামতের আগে আর মেয়েটাকে দেখতে পাবো না'	বাংলাদেশে তাজরীন গর্মেটসে ভয়াবহ অরিকাতের সাত বছর পরও রাষ্ট্রপক্ষ শুনানিতে সাক্ষী হাতির করতে পারছে না বলে মামলার অগ্রগতি নেই। ২০১২ সালের ২৪ নভেম্বর সন্ধ্যায় ভয়াবহ সেই অরিকাতে ১২ জন শ্রমিক প্রাণ হারিয়েছিলেন। সেই ঘটনায় তরা মামলায় কয়েকদিন পর প...	BBC Bangla	11/24/2019	Govt critique
4	শাহবাগ আন্দোলন	শাহবাগের সেই 'পপজাগরণ মঞ্চ' এখন কোথায়, কী করছে?	বাংলাদেশে ১৯৭১ সালের মানবতাবিরোধী অপরাধীদের মুক্তাদেওর দাবিতে গড়ে ওঠা 'পপজাগরণ মঞ্চ' পাঁচ বছর পূর্তি হলো আজ। মানবতাবিরোধী অপরাধের অভিযোগে ২০১৩ সালের ৫ ফেব্রুয়ারি আন্তর্জাতিক অপরাধ ট্রাইব্যুনাল আদালত কাদের মোল্লাকে যাবজ্জীবন কারাদণ্ড দেন। এর পর একদল তরুণ...	Jugantor	2/5/2018	Neutral
6	হেফাজতে ইসলামের নংমাট ও ঢাকা অবরোধ	হেফাজতে নেতাদের নজরদারির পরামর্শ একদফা আন্দোলনে নামানোর পরিকল্পনা চলছে - হাটহাজারী মাদ্রাসায় বিশেষ দৃষ্টি...	আগামী দ্বাদশ জাতীয় সংসদ নির্বাচনকে ঘিরে হেফাজতে ইসলামের শীর্ষ নেতাদের গতিবিধিসহ সব কার্যক্রম সরকারের নজরদারিতে আনার পরামর্শ দিয়েছে গোয়েশারা। অরাজনৈতিক এ সংগঠন যেন বিনোদন-জন্মায়তের দিকে ঝুঁকিত হতে না পারে, সে বিষয়টি পর্যালোচনা রাখা উচিত বলেও মনে করেন গোয়েশা...	Jugantor	9/23/2023	Govt leaning

The raw corpus stores complete article text and an article-level framing label. Sentence segmentation and six-aspect labels are added in the annotated dataset.

Figure 1: Sample of the base BanglaBias article dataset obtained from Hugging Face [3]. It contains document metadata, headline and body text, source information, and one article-level stance label.

3.2 Project-specific sentence and aspect annotation

The original BanglaBias article-level labels provide useful document-level supervision, but they are not sufficient for the more fine-grained objective of this project. Our system needs to identify *what governmental aspect* a particular sentence discusses and *how the government is framed with respect to that aspect*. We therefore constructed an additional sentence-level annotation layer from a subset of the BanglaBias base corpus.

The completed project annotation file contains 5,308 sentence rows derived from 118 of the base articles and covering 26 events. There are 118 headline rows, 118 lead rows, and 5,072 body-sentence rows. Each sentence retains its original article and section metadata so that it remains traceable to the source article in the base dataset.

For the project-specific annotation, each sentence was evaluated separately against six governmental aspects. This adds six stance annotations, one for each aspect, using the labels NR, N, GF, and GC. The annotation file additionally records `stance_holder` and `is_quote`, allowing quoted or attributed framing to be distinguished during analysis.

Thus, the relationship between the two datasets can be summarised as follows: the Hugging Face BanglaBias dataset is the **base article corpus**, while the sentence-level, six-aspect annotation dataset is a **project-specific extension derived from a subset of that corpus**. The article text and original article-level labels originate from BanglaBias; the aspect-specific sentence annotations and associated sentence-level metadata used for the two-stage classification task were added later for this project.

Figure 2 intentionally focuses on these project-specific annotation columns. The six central columns, A1–A6, contain one of NR, N, GF, or GC. The two rightmost highlighted fields record who holds the stance and whether the text is a quotation. The colours in the screenshot are explanatory rendering only; the stored values remain categorical text.

Annotated Sentence Dataset — Annotation Columns in Focus									
Each sentence has six target-aware aspect labels plus attribution metadata									
sentence_id	sentence_text	A1	A2	A3	A4	A5	A6	stance_holder	is_quote
38_H	রামপাল বিদ্যুৎকেন্দ্রে সুন্দরবনের ক্ষতি হবে না: প্রধানমন্ত্রী	GF	NR	NR	NR	GF	GF	journalist	0
55_S051	এর ফলে রাষ্ট্রীয় ব্যয় বাড়ছে	GC	NR	NR	NR	GC	NR	journalist	0
26_S022	আহত ২৫০ জন।	NR	NR	N	NR	NR	NR	journalist	0
108_S014	ভারা তুধু যাদের ভিন্নমত তাদের দমন করতে পারে।	NR	NR	GC	NR	NR	NR	journalist	0

Annotation labels: NR — not relevant N — neutral GF — government-favourable GC — government-critical

A1–A6 are the supervised targets. The same sentence may be relevant to multiple aspects with different labels. stance_holder and is_quote support attribution and error analysis but are not model inputs.

Figure 2: Project-specific sentence-level annotations derived from a subset of the BanglaBias base corpus. The six aspect columns were added for this project. Grey denotes NR, blue N, green GF, and red GC. The `stance_holder` and `is_quote` fields provide additional attribution context.

Expanding every annotated sentence across six aspects produces 31,848 sentence–aspect pairs. Table 2 exposes the central difficulty of the project: relevant examples form a minority, and government-favouring and government-critical pairs are extremely rare.

Table 2: Pair-level label distribution in the project-specific annotated dataset.

Label	Count	Percentage	Role
NR	24,630	77.34%	Stage 1 negative class
N	6,882	21.61%	Stage 1 positive; Stage 2 class
GF	108	0.34%	Stage 1 positive; Stage 2 class
GC	228	0.72%	Stage 1 positive; Stage 2 class
Total	31,848	100.00%	Six pairs per sentence

3.3 Cleaning and preprocessing evidence

Preprocessing was explicitly performed and checked. The process normalises whitespace and selected Unicode variants in a separate cleaned-text field while preserving the original sentence field for traceability. It removes neither article rows nor valid sentence rows. Empty cleaned strings are detected, reviewed, and safely handled instead of silently entering training. The transformation then:

1. reconstructs headline, lead, and body sentence order for each article;
2. validates the permitted annotation values in all six aspect columns;
3. converts each sentence into six sentence–aspect pairs;
4. maps pair labels to binary relevance targets and conditional three-class stance targets;
5. tokenises the sentence and full aspect description as a paired sequence; and
6. stores the processed pairs and deterministic split assignments as reusable artifacts.

Token analysis found a median paired length of 28 tokens, a 95th percentile of 51, a 99th percentile of 77, and a maximum of 221. With project maximum length 128, only 53 of 31,848 pairs (0.17%) require truncation. This retains almost all content while reducing memory and training time relative to BanglaBERT’s architectural maximum of 512 positions.

4 Data Splitting and Leakage Control

Random row splitting would leak event-specific names, entities, and wording because several articles and many sentences describe the same event. The project therefore uses a deterministic event-grouped split with seed 42: 18 events for training, four for validation, and four for testing. This corresponds to 83/17/18 articles and 21,570/3,930/6,348 pairs for train/validation/test respectively.

The four protected test events concern the Rampal coal power agreement, the BNP and Awami League grand rallies, the Nusrat murder case, and the casino enforcement campaign. No sentence from these events is used in either transformer training. This is stricter and more realistic than a random sentence split because it measures transfer to unseen stories.

5 System Architecture

The language encoder is BanglaBERT, an ELECTRA-style discriminator pretrained specifically for Bangla [4, 5]. Contextual transformers build token representations using bidirectional context, following the broad paradigm established by BERT [6]. The implementation uses the Hugging Face Transformers training and inference stack [7, 8].

For a sentence s and aspect description a_k , the tokenizer forms

$$[\text{CLS}] \ s \ [\text{SEP}] \ a_k \ [\text{SEP}], \quad (1)$$

with padding or truncation to 128 tokens. BanglaBERT has a 32,000-token vocabulary, hidden size 768, 12 transformer layers, and 12 attention heads. Thus a batch of B paired sequences enters as token IDs of shape $B \times 128$ and becomes contextual embeddings of shape $B \times 128 \times 768$. The contextual [CLS] vector has shape $B \times 768$ and feeds the task-specific classification head.

Figure 3 gives the complete architecture, including tensor dimensions and the connection between sentence-level inference and the final user-facing result.

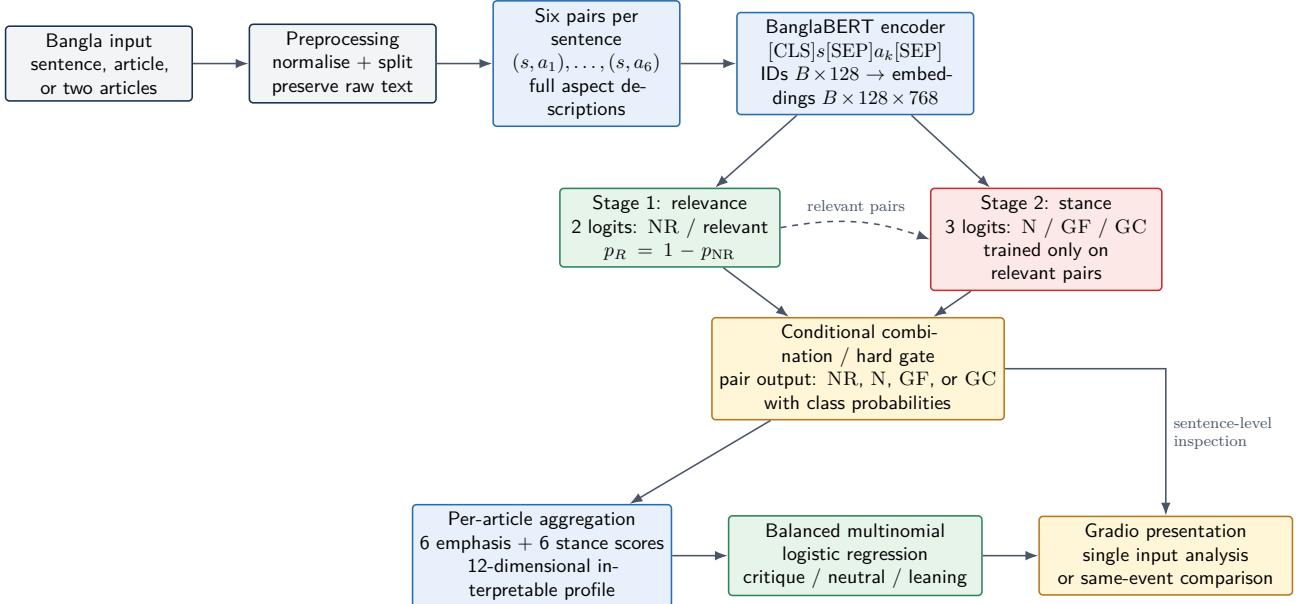


Figure 3: Detailed end-to-end architecture. Dashed flow indicates that stance is conceptually conditional on relevance; the solid paths show the probabilities combined by the application.

5.1 Stage 1: aspect relevance

Stage 1 is a two-class BanglaBERT sequence classifier. Its training target maps N, GF, and GC to relevant and retains NR as not relevant. At inference, it returns $p_R(s, a_k)$ and $p_{NR}(s, a_k)$. This model prevents irrelevant aspects from being assigned misleading stance labels.

5.2 Stage 2: conditional stance

Stage 2 is a separate three-class BanglaBERT sequence classifier trained only on gold-relevant pairs. It estimates

$$\mathbf{p}_S(s, a_k) = [p_N, p_{GF}, p_{GC}]. \quad (2)$$

Class-weighted cross-entropy is used to reduce dominance by N. If n_c is the training count for class c , the explored weights follow

$$w_c \propto \left(\frac{1}{n_c}\right)^p, \quad (3)$$

where the original run used inverse-square-root weighting ($p = 0.50$) and a later experiment used $p = 0.75$.

At inference, the relevance threshold supplies the explicit gate. If a pair is not relevant, its output is NR. Otherwise, the largest conditional stance probability supplies N, GF, or GC. The interface also preserves probabilities instead of showing only the hard class, enabling uncertainty inspection.

5.3 Article aggregation and final classifier

Suppose article d has m sentences. For aspect a_k , its emphasis feature is the mean predicted relevance:

$$E_k(d) = \frac{1}{m} \sum_{i=1}^m p_R(s_i, a_k). \quad (4)$$

Its directional stance feature is a relevance-weighted mean of favouring minus critical probability:

$$S_k(d) = \frac{\sum_{i=1}^m p_R(s_i, a_k) [p_{GF}(s_i, a_k) - p_{GC}(s_i, a_k)]}{\sum_{i=1}^m p_R(s_i, a_k) + \epsilon}. \quad (5)$$

The article is therefore represented by

$$\mathbf{x}_d = [E_1, \dots, E_6, S_1, \dots, S_6] \in \mathbb{R}^{12}. \quad (6)$$

A multinomial logistic-regression classifier with balanced class weights maps this profile to *Govt critique*, *Neutral*, or *Govt leaning*. The regularisation parameter C was selected from $\{0.01, 0.1, 1, 10, 100\}$ using grouped cross-validation, with $C = 100$ selected. Logistic regression was chosen because the article dataset is small and its feature coefficients remain inspectable; the implementation uses scikit-learn [9].

6 Complete Input-to-Output Workflow

This section follows one input all the way to the final result.

1. **Input and reconstruction.** The user enters a Bangla sentence, a full article, or two articles about the same event. For a complete article, the application segments text into ordered sentences. The raw input remains available for display.
2. **Text normalisation.** Whitespace and selected Unicode inconsistencies are normalised using the same rules as training. Empty segments are discarded only after validation.

3. **Aspect expansion.** Every sentence is duplicated six times and paired with the six full aspect descriptions. An article of m sentences therefore creates $6m$ model inputs.
4. **Tokenisation.** Each pair is converted into token IDs, attention masks, and token-type IDs using the saved BanglaBERT tokenizer. The sequence is padded or truncated to 128 positions.
5. **Relevance inference.** Stage 1 returns NR/relevant logits, which softmax converts to probabilities. The configurable threshold, 0.50 by default, determines the relevance gate.
6. **Stance inference.** Stage 2 returns N/GF/GC probabilities. These are interpreted only when Stage 1 marks the pair relevant.
7. **Sentence display.** For a single sentence, the UI shows all six aspect results, not merely one overall label. This reveals which aspect activated, its relevance confidence, and its conditional stance.
8. **Article profile.** For a full article, probabilities are aggregated using E_k and S_k , producing six emphasis values and six stance values.
9. **Final article decision.** The saved logistic-regression model converts the 12-dimensional profile into three article probabilities and a final critique/neutral/leaning label.
10. **Two-article comparison.** When two same-event articles are supplied, Steps 1–9 run independently for both. The UI aligns their final probabilities and aspect profiles, showing where their emphasis or governmental direction differs.

This design preserves a traceable path from the final label back to sentence–aspect evidence. For example, a government-critical article result can be examined to determine whether it arose mainly from accountability, security, economic, or welfare sentences rather than from an unexplained black-box score.

7 Training Configuration and Evaluation

All transformer checkpoints use BanglaBERT [4]. Training uses a fixed seed, grouped splits, dynamic batching, evaluation at epoch boundaries, and metric-based best-model selection. The later experimental notebook adds explicit early stopping. Accuracy is reported for intuitive correctness, while macro F1 is the main model-selection statistic because it gives equal importance to common and rare labels. Weighted F1 is included for the end-to-end system but can conceal minority-class failure.

The evaluation is separated into four views:

- Stage 1 on all protected test pairs;
- Stage 2 on gold-relevant protected test pairs;
- the end-to-end four-label output on all protected test pairs; and
- the article classifier on 18 held-out articles from unseen events.

8 Results

8.1 Stage 1 relevance results

Stage 1 is strong and well balanced compared with the later stance problem. Table 3 shows 0.9631 accuracy, 0.9457 macro F1, and 0.9577 relevant-class precision–recall AUC.

Table 3: Stage 1 protected test performance (6,348 pairs).

Class / summary	Precision	Recall	F1	Support
NR	0.9729	0.9800	0.9765	4,953
Relevant	0.9272	0.9032	0.9150	1,395
Accuracy		0.9631		6,348
Macro F1		0.9457		6,348
Relevant PR-AUC		0.9577		6,348

8.2 Stage 2 stance results

Stage 2 correctly handles the large neutral class but struggles with directional framing. Its 0.9348 accuracy appears high because 1,314 of 1,395 relevant test pairs are neutral. Macro F1 gives the more honest summary of 0.5098. The model still improves substantially over the always-neutral macro F1 of 0.3234 and detects some rare GF and GC cases.

Table 4: Selected Stage 2 performance on gold-relevant protected test pairs.

Class / summary	Precision	Recall	F1	Support
N	0.9555	0.9795	0.9673	1,314
GF	0.4583	0.3548	0.4000	31
GC	0.2500	0.1200	0.1622	50
Accuracy		0.9348		1,395
Macro F1		0.5098		1,395
Weighted F1		0.9258		1,395

8.3 End-to-end pair and article results

When both transformer stages are connected, the final four-label sentence–aspect system achieves 0.9504 accuracy, 0.6060 macro F1, and 0.9484 weighted F1 over all 6,348 test pairs. The error decomposition is especially informative: 4,854 pairs are correctly rejected as NR; 99 are Stage 1 false positives; 135 are Stage 1 false negatives; 1,179 are correct through both relevance and stance; and 81 have correct relevance but incorrect stance. Thus, further gains now depend more on rare stance recognition than on relevance.

At article level, the 12-feature balanced logistic regression obtains 0.4444 accuracy and 0.4254 macro F1 on 18 held-out articles, compared with 0.2778 accuracy and 0.1449 macro F1 for an always-neutral classifier. Government-leaning performance is strongest (F1 0.7500), government-critique reaches F1 0.5263, and neutral recall is 0. The result is above the majority baseline but is not yet robust enough for high-stakes deployment; only 118 article profiles were available, and the neutral test support is five.

Table 5: Summary of the selected final pipeline.

Evaluation	Accuracy	Macro F1	Weighted F1	Size	Main interpretation
Stage 1 relevance	0.9631	0.9457	–	6,348 pairs	Strong relevance filter across unseen events.
Stage 2 stance	0.9348	0.5098	0.9258	1,395 pairs	High accuracy is dominated by neutral; minority stance remains difficult.
End-to-end four label	0.9504	0.6060	0.9484	6,348 pairs	Best measure of the deployed sentence–aspect pipeline.
Article classifier	0.4444	0.4254	0.4298	18 articles	Better than majority baseline, but neutral articles are not recovered.

9 Experiments to Improve Minority-Class Accuracy

9.1 Why accuracy alone was insufficient

Only 108 GF and 228 GC examples exist in all 31,848 pairs, compared with 6,882 neutral relevant pairs. A classifier can therefore obtain high accuracy while missing most directional sentences. Every experiment was judged primarily by macro F1 and per-class recall, and was ultimately checked against original-news event-held-out data.

9.2 Experiment 1: inverse-square-root class weighting

The first Stage 2 model used class weights with exponent $p = 0.50$. Validation accuracy reached 0.9637 and macro F1 0.3986. Neutral F1 was 0.9814; GF F1 was 0; and GC precision/recall/F1 were 0.7500/0.1250/0.2143. Validation loss began increasing after the first epoch even while training continued, indicating overfitting. The original notebook did not include early stopping, which was a weakness identified during analysis.

9.3 Experiment 2: stronger class weighting and early stopping

Increasing the exponent to $p = 0.75$ penalised minority mistakes more heavily. Validation accuracy became 0.9649 and macro F1 0.4014. GC precision rose to 1.0, but recall remained only 0.125 and GF F1 remained 0. The macro-F1 improvement was approximately 0.0028, too small to justify replacing the selected model. This experiment demonstrated that loss manipulation cannot create missing linguistic diversity: it changes the cost of mistakes but not the number or variety of rare examples.

The revised training configuration introduced early stopping, metric-based checkpoint selection, and retention of only the best checkpoint. Early stopping is appropriate because validation loss showed that later epochs increasingly memorised the training distribution. It also reduces storage use: repeated transformer checkpoints can consume several gigabytes on Google Drive or Kaggle.

9.4 Experiment 3: corrected, exactly balanced Stage 2 dataset

A separate Stage 2-only experiment used a corrected, exactly balanced dataset of 2,940 relevant pairs: 980 each for N, GF, and GC. Its split contained 2,193 training rows, 519 selection-validation rows, 94 reference-validation rows, and 134 protected-test rows according to the experimental bookkeeping. Early stopping halted after epoch 7. Validation loss reached its minimum at epoch 3 (0.1989), then increased through epochs 4–6 (0.2195, 0.2625, and 0.2898) while macro F1 plateaued around 0.92. This again indicated overfitting.

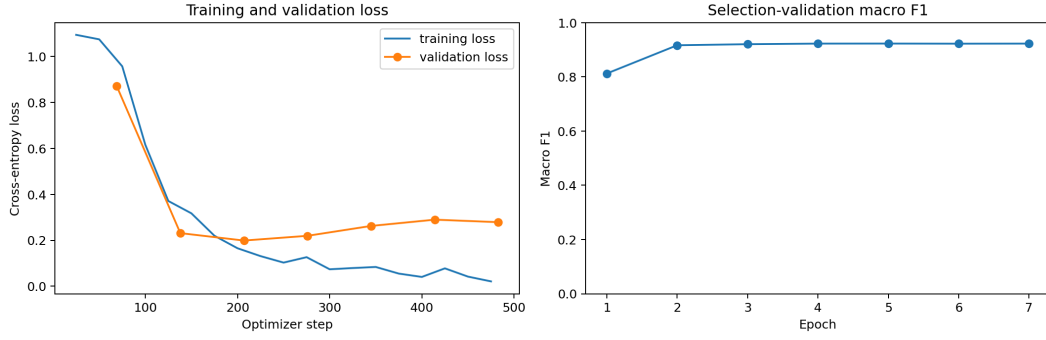


Figure 4: Training history from the balanced Stage 2 experiment. The divergence between training improvement and validation loss motivated early stopping and evaluation on untouched original-news data.

The balanced model appeared excellent on its internal selection validation split: 0.9306 accuracy and 0.9224 macro F1, with class F1 values of 0.8800 for N, 0.9976 for GF, and 0.8896 for GC. However, a style/origin audit revealed that macro F1 was 1.0000 on 371 generated-style rows but only 0.3887 on 148 original-news rows. On the separate original reference validation data it reached only 0.4387 macro F1. This discrepancy showed that the internal balanced score measured a substantially easier distribution.

The decisive comparison used the exact same 1,395 protected original-news relevant pairs as the selected model. Table 6 shows that balancing increased GF recall from 0.3548 to 0.4194, but sharply reduced precision and overall reliability. GC recall increased only from 0.12 to 0.14 while precision collapsed from 0.25 to 0.0583. The balanced model therefore generated many false directional alarms.

Table 6: Selected and balanced Stage 2 models on the same protected original-news test set.

Model	Accuracy	Macro F1	Weighted F1	N F1	GF F1	GC F1
Selected Stage 2	0.9348	0.5098	0.9258	0.9673	0.4000	0.1622
Balanced Stage 2	0.8480	0.4375	0.8735	0.9168	0.3133	0.0824

Replacing the selected stance model also lowered complete four-label performance from 0.9504 accuracy / 0.6060 macro F1 to 0.9321 / 0.5455. Event-wise, the balanced model improved the Nusrat and grand-rally events but worsened Rampal and casino-event performance; only A1 improved while A2–A6 declined. The balanced checkpoint was therefore retained as an experiment and not used as a replacement.

9.5 Experiment conclusions

1. Class weighting is useful but cannot compensate for extremely limited GF/GC language.
2. Exact numeric balance does not guarantee representative linguistic balance.
3. Internal validation can be misleading when writing style or generation process differs from real news.
4. Event-held-out original-news evaluation is necessary before replacing a selected checkpoint.
5. Future improvement should prioritise diverse, independently reviewed directional examples from many events and outlets, followed by a newly reserved event-held-out test set.

10 User Interface

The local interface is implemented with Gradio [10] and loads the two transformer directories, the tokenizer assets, the article classifier, and experiment configuration from the project artifacts. It runs

on CPU or GPU and reports the detected device. Figure 5 is an actual screenshot of the running application, not a mock-up.

The *Single Sentence* tab accepts a Bangla sentence and reports all six aspect decisions and scores. The *Compare Two Articles* tab accepts two same-event reports and produces side-by-side article labels, confidence values, aspect emphasis, and stance differences. A *Model Information* tab explains label meanings and the saved runtime configuration. The default 0.50 relevance threshold is visible and adjustable for demonstrations, making the relationship between thresholding and false positives observable.

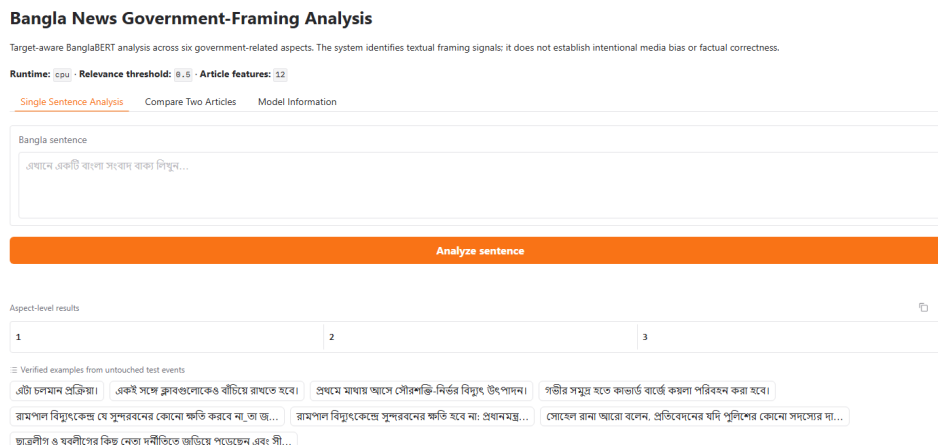


Figure 5: Actual local Gradio interface using the saved final artifacts. The tabs support one-sentence analysis, two-article comparison, and model information.

For classroom use, results should be described as model predictions. A very high neutral score for an intuitively critical sentence may occur because directional examples were scarce in the training data and because criticism can be expressed through omission, negation, or context-dependent language that was rare during training. In the protected test set, only 50 government-critical pairs were available, further illustrating the scarcity of this class. Showing all six aspect probabilities and, for articles, the underlying sentence evidence prevents the final label from being presented without context.

11 Key Findings and Discussion

The project produced six main findings.

First, decomposing relevance from stance was effective. Stage 1 reached 0.9457 macro F1 and removed most irrelevant pairs before stance interpretation. Second, raw accuracy is not an adequate success criterion under severe imbalance; the 0.9348 Stage 2 accuracy coexists with a GC F1 of only 0.1622. Third, the end-to-end error flow confirms that relevance is no longer the primary bottleneck: stance errors and Stage 1 false negatives among rare directional content are more consequential.

Fourth, stronger loss weights produced almost no meaningful gain. Weighting changes decision pressure, but a model still needs diverse examples of explicit criticism, implicit criticism, praise, attribution, quotation, negation, and context-dependent language. Fifth, the balanced experiment illustrated a general machine-learning risk: a perfectly balanced dataset may yield impressive internal validation scores while transferring poorly to authentic target-domain writing. The protected test comparison prevented a harmful model replacement.

Finally, the article classifier shows promise but remains the least certain component. Twelve interpretable aggregate features are preferable to an opaque document vector for this small dataset, yet 118 annotated articles covering 26 events cannot capture the full variation of the 200-article, 46-event base corpus. The

zero neutral recall on the 18-article test set should be reported plainly. More event-diverse articles and nested or out-of-fold transformer features are needed before claiming robust article-level generalisation.

12 Limitations, Ethics, and Future Work

The project builds on the publicly available BanglaBias article corpus [2, 3]; however, the sentence-level aspect annotation scheme introduced in this project is distinct from the original BanglaBias article-level benchmark. The project-specific sentence annotations are not part of the original BanglaBias release and should not be interpreted as annotations produced or validated by its original authors.

The sentence annotations used here are project annotations rather than a large independently adjudicated human gold standard. Inter-annotator agreement is therefore unavailable. Directional labels are scarce, quoted speech can be mistaken for the outlet’s own stance, and sentence-only inputs may omit context required to resolve agency or negation. The final article stack is not fully cross-fitted, and the held-out article sample is small.

The tool must not be used to assert that a journalist or organisation intentionally supports or opposes the government. It detects patterns learned from this label scheme. Predictions should be paired with probabilities, sentence evidence, and an explicit uncertainty statement. Misclassification risk is highest for rare directional classes and context-dependent statements.

Recommended next work is:

1. collect more authentic GF and GC examples across additional events, sources, and linguistic constructions;
2. perform independent double annotation and report agreement and adjudication procedures;
3. reserve a new untouched event-level test set before further tuning, because the current protected test has now informed model comparison;
4. explore focal loss, calibrated thresholds, and constrained joint probabilities only after data expansion;
5. model surrounding-sentence context and explicitly incorporate quotation/stance-holder metadata;
6. generate article features out-of-fold to remove stacking optimism; and
7. calibrate probability outputs and add confidence-based abstention for uncertain classroom examples.

13 Reproducibility and Deliverables

The original article corpus used as the starting point of this project is publicly accessible through the BanglaBias Hugging Face repository [3]. This makes the base article data independently identifiable and reproducible. The project then adds a separate sentence-level processing and annotation layer required for the six-aspect relevance-and-stance task.

The main training notebook records preprocessing, pair construction, transformer training, evaluation, article aggregation, and artifact export. A separate Stage 2-only notebook records the balanced-data experiment and its early-stopping history. The deployable system requires the source application, both saved transformer model directories, tokenizer files, the article classifier, the aspect/config metadata, and Python dependencies. The generated report assets are derived directly from the base and project-specific datasets or the running UI.

The principal reusable outputs are:

- the BanglaBias base article corpus obtained from Hugging Face;
- the project-specific sentence-level and aspect-level annotation dataset derived from a subset of the base corpus;
- `artifacts/relevance_model/`: Stage 1 model and tokenizer;
- `artifacts/stance_model/`: selected final Stage 2 model and tokenizer;
- `artifacts/article_classifier.joblib`: article-level classifier;

- processed pairs, split assignments, metrics, and prediction files for auditing;
- the local Gradio application for interactive inference; and
- this LaTeX source and compiled PDF report.

14 Conclusion

The project accomplishes its task through a transparent hierarchy: sentence–aspect relevance, conditional sentence–aspect stance, interpretable article aggregation, and final article classification. Starting from the publicly available BanglaBias article corpus, the project extends the data representation through project-specific sentence-level annotation across six governmental aspects. This extension transforms the original article-level stance resource into a finer-grained dataset suitable for analysing both the aspect being discussed and the stance expressed toward the government within individual sentences.

Its strongest component is relevance detection; its main unresolved challenge is robust recognition of rare favourable and critical framing. The selected final pipeline achieves 0.6060 macro F1 across the complete four-label pair task and provides useful explanatory aspect profiles, while the article-level result remains preliminary. Careful experiments showed that neither stronger class weighting nor a numerically balanced alternative dataset should replace the selected stance model. Most importantly, the workflow, interface, and evaluation expose these strengths and weaknesses rather than hiding them behind aggregate accuracy.

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